📚 MANUAL BOOK WEBSITE PROFILE SMKN 5 SAMARINDA

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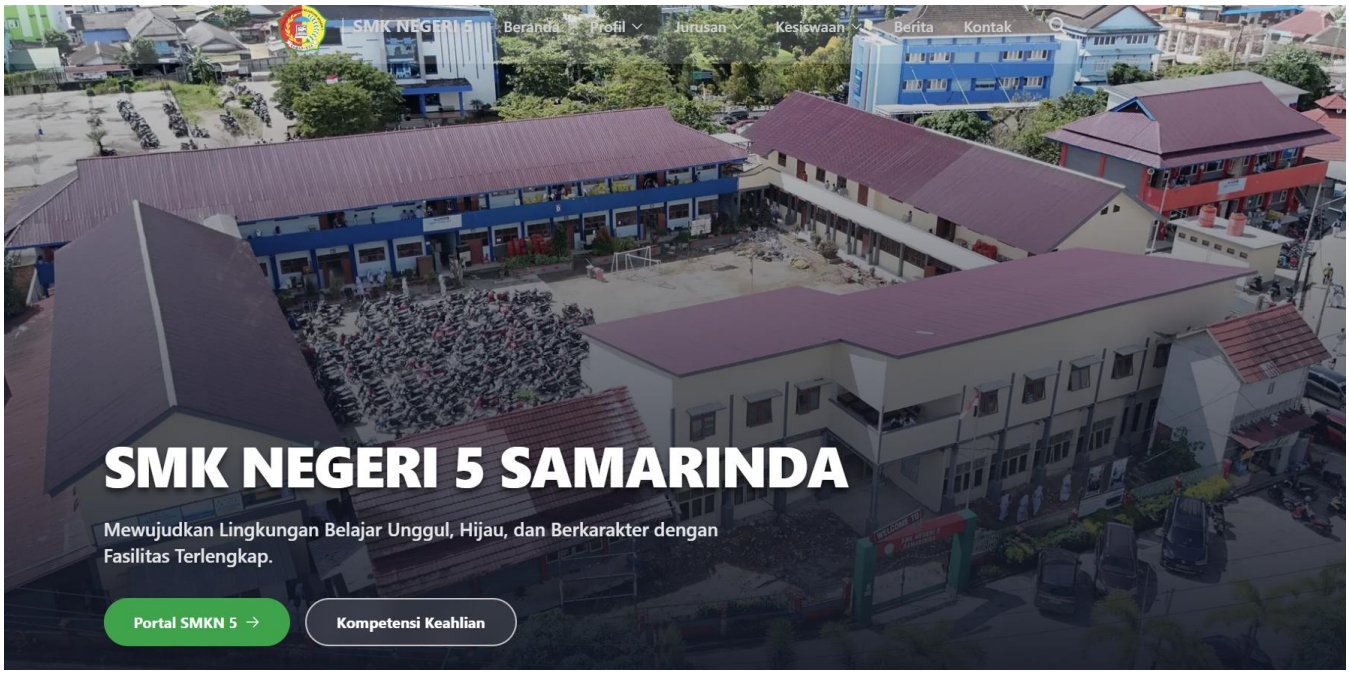
1. RINGKASAN SISTEM

1.1 Tentang Aplikasi

Website Profile SMK Negeri 5 Samarinda (https://smkn5smr.sch.id) adalah sistem informasi berbasis web yang dirancang untuk:

- **Menampilkan informasi sekolah** (profil, visi-misi, sejarah, struktur organisasi)
- **Publikasi berita** dengan sistem role-based workflow (Jurnalis → Redaktur → Publikasi)
- **Informasi program keahlian** (jurusan) dengan detail lengkap
- **Informasi ekstrakurikuler** dan fasilitas sekolah
- **Panel administrasi modern** menggunakan Filament 4
- **Responsive design** dengan Tailwind CSS 4

1.2 Tampilan Website (https://smkn5smr.sch.id)







1.3 Karakteristik Sistem

Aspek	Detail
Bahasa Pemrograman	PHP 8.2+

Aspek	Detail
Framework	Laravel 12.0
Admin Panel	Filament 4.3
Database	SQLite (Development), MySQL/PostgreSQL (Production)
Frontend	Blade Templates + Alpine.js 3.15
CSS Framework	Tailwind CSS 4.0
Build Tool	Vite 7.0
Permission	Spatie Laravel Permission 6.24
Server Requirement	PHP 8.2, Composer, Node.js 18+

2. ARSITEKTUR & TEKNOLOGI

2.1 Stack Teknologi

Backend Dependencies (composer.json)

```
{
  "php": "^8.2",
  "filament/filament": "^4.3",
  "laravel/framework": "^12.0",
  "laravel/tinker": "^2.10.1",
  "spatie/laravel-permission":  "^6.24"
}
```

Frontend Dependencies (package.json)

```
{
  "dependencies": {
    "alpinejs": "^3.15.3"
  },
  "devDependencies": {
    "@tailwindcss/vite": "^4.0.0",
    "axios": "^1.11.0",
    "laravel-vite-plugin":  "^2.0.0",
    "tailwindcss": "^4.0.0",
    "vite": "^7.0.7"
  }
}
```

2.2 Pola Arsitektur



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3. STRUKTUR FOLDER

3.1 Struktur Utama

```
Profile-SMKN5/
│
├── app/
│   ├── Filament/                    # Admin Panel Resources
│   │   ├── Resources/               # CRUD Resources
│   │   │   ├── Beritas/            # Berita Resource
│   │   │   ├── Jurusans/           # Jurusan Resource
│   │   │   ├── Users/              # User Management
│   │   │   ├── Settings/           # Website Settings
```

```
│   │   │   ├── Galleries/          # Gallery Management
│   │   │   ├── Facilities/         # Facility Management
│   │   │   ├── Partners/           # Partner/Mitra
│   │   │   ├── Menus/              # Dynamic Menu
│   │   │   └── Extracurriculars/   # Ekstrakurikuler
│   │   ├── Widgets/                # Dashboard Widgets
│   │   │   └── StatsOverview.php   # Statistics Widget
│   │   └── Providers/
│   │       └── AdminPanelProvider.php  # Panel Configuration
│   │
│   ├── Http/
│   │   └── Controllers/
│   │       ├── BeritaController.php      # Berita Frontend
│   │       ├── HomeController.php        # Homepage
│   │       ├── JurusanController.php     # Jurusan Frontend
│   │       └── ProfileController.php     # Profil Pages
│   │
│   ├── Models/                     # Eloquent Models
│   │   ├── User.php                # User (Auth + Roles)
│   │   ├── Berita.php              # News/Berita
│   │   ├── Jurusan.php             # Program Keahlian
│   │   ├── Setting.php             # Site Settings
│   │   ├── Gallery.php             # Gallery Albums
│   │   ├── GalleryImage.php        # Gallery Images
│   │   ├── Facility.php            # Facilities
│   │   ├── Partner.php             # Partners/Mitra
│   │   ├── Menu.php                # Dynamic Navigation
│   │   ├── Statistic.php           # Site Statistics
│   │   ├── Extracurricular.php     # Ekstrakurikuler
│   │   └── Agenda.php              # School Agenda/Events
│   │
│   ├── Policies/                   # Authorization Policies
│   │   ├── BeritaPolicy.php        # Berita Access Control
│   │   └── UserPolicy.php          # User Management Access
│   │
│   └── Providers/
│       └── AppServiceProvider.php  # Global Service Provider
│
├── bootstrap/
│   ├── app.php                     # Application Bootstrap
│   └── providers.php               # Service Providers Config
│
├── config/                         # Configuration Files
│   ├── app.php                     # Application Config
│   ├── auth.php                    # Authentication Config
│   ├── database.php                # Database Config
│   ├── filesystems.php             # File Storage Config
│   ├── permission.php              # Spatie Permission Config
│   └── session.php                 # Session Config
│
├── database/
│   ├── migrations/                 # Database Migrations
│   │   ├── 0001_01_01_000000_create_users_table.php
│   │   ├── 2025_12_30_120934_create_jurusan_table.php
│   │   ├── 2025_12_30_120958_create_berita_table.php
│   │   ├── 2026_01_06_141315_create_permission_tables.php
```

```
│   │   └── ... (18 migration files total)
│   │
│   ├── seeders/                      # Database Seeders
│   │   ├── DatabaseSeeder.php        # Master Seeder
│   │   ├── RolePermissionSeeder.php  # Roles & Permissions
│   │   ├── UserRoleSeeder.php        # Default Users
│   │   ├── JurusanSeeder.php         # Sample Jurusan
│   │   ├── BeritaSeeder.php          # Sample News
│   │   ├── StatisticSeeder.php       # Site Statistics
│   │   ├── MenuSeeder.php            # Navigation Menu
│   │   ├── SettingSeeder.php         # Site Settings
│   │   └── AgendaSeeder.php          # School Events
│   │
│   └── factories/
│       └── UserFactory.php           # User Factory
│
├── public/
│   ├── index.php                     # Application Entry Point
│   ├── assets/                       # Static Assets
│   │   └── images/                   # Images
│   │       └── pimpinan/             # Leadership Photos
│   ├── images/                       # Public Images
│   │   ├── logo-smk-utama.png       # Main Logo
│   │   └── drone-smk-1.webp         # Hero Images
│   ├── storage/                      # Symbolic Link to Storage
│   └── build/                        # Vite Build Output
│
├── resources/
│   ├── css/
│   │   └── app.css                   # Main Stylesheet
│   ├── js/
│   │   └── app.js                    # Main JavaScript
│   └── views/                        # Blade Templates
│       ├── layouts/
│       │   └── app.blade.php         # Main Layout
│       ├── components/               # Blade Components
│       │   ├── navbar.blade.php      # Navigation Bar
│       │   └── footer.blade.php      # Footer
│       ├── home.blade.php            # Homepage
│       ├── berita/                   # News Pages
│       │   ├── index.blade.php       # News List
│       │   └── detail.blade.php      # News Detail
│       ├── jurusan/                  # Jurusan Pages
│       │   ├── index.blade.php       # Jurusan List
│       │   └── detail.blade.php      # Jurusan Detail
│       ├── profil/                   # Profile Pages
│       │   ├── tentang.blade.php     # About School
│       │   ├── unit.blade.php        # School Units
│       │   └── struktur.blade.php    # Organization Structure
│       └── kesiswaan/                # Student Affairs
│           └── ekskul.blade.php      # Extracurricular
│
├── routes/
│   ├── web.php                       # Web Routes
│   └── console.php                   # Console Commands
│
```

```
├── storage/
│   ├── app/                        # Application Storage
│   │   ├── public/                 # Public Accessible Files
│   │   └── private/                # Private Files
│   ├── framework/                  # Framework Storage
│   └── logs/                       # Application Logs
│
├── tests/                          # Unit & Feature Tests
│
├── .env.example                    # Environment Template
├── composer.json                   # PHP Dependencies
├── package.json                    # NPM Dependencies
├── vite.config.js                  # Vite Configuration
├── artisan                         # Artisan CLI
└── README.md                       # Project Documentation
```

---

4. KONFIGURASI ENVIRONMENT

4.1 File .env (Environment Variables)

File `.env` berisi konfigurasi sensitif dan environment-specific. **JANGAN** commit file ini ke repository!

Template .env untuk Development

```
# Application Settings
APP_NAME="SMKN 5 Samarinda"
APP_ENV=local
APP_KEY=base64:xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
APP_DEBUG=true
APP_URL=http://localhost

# Localization
APP_LOCALE=id
APP_FALLBACK_LOCALE=id
APP_FAKER_LOCALE=id_ID

# Database (Development - SQLite)
DB_CONNECTION=sqlite
# DB_DATABASE akan otomatis ke database/database.sqlite

# Session & Cache
SESSION_DRIVER=database
SESSION_LIFETIME=120
CACHE_STORE=database
QUEUE_CONNECTION=database

# Mail Configuration (Development - Log Only)
MAIL_MAILER=log
MAIL_FROM_ADDRESS="noreply@smkn5samarinda.sch.id"
MAIL_FROM_NAME="${APP_NAME}"

# File Storage
```

```
FILESYSTEM_DISK=public

# Logging
LOG_CHANNEL=stack
LOG_LEVEL=debug
```

Template .env untuk Production

```
# Application Settings
APP_NAME="SMKN 5 Samarinda"
APP_ENV=production
APP_KEY=base64:xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
APP_DEBUG=false
APP_URL=https://smkn5samarinda.sch.id

# Localization
APP_LOCALE=id
APP_FALLBACK_LOCALE=id

# Database (Production - MySQL)
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=smkn5_profile
DB_USERNAME=smkn5_user
DB_PASSWORD=StrongPassword123!

# Session & Cache
SESSION_DRIVER=database
SESSION_LIFETIME=120
SESSION_SECURE_COOKIE=true
CACHE_STORE=database

# Mail Configuration (Production - SMTP)
MAIL_MAILER=smtp
MAIL_HOST=smtp.gmail.com
MAIL_PORT=587
MAIL_USERNAME=your-email@gmail.com
MAIL_PASSWORD=your-app-password
MAIL_ENCRYPTION=tls
MAIL_FROM_ADDRESS="noreply@smkn5samarinda.sch.id"
MAIL_FROM_NAME="${APP_NAME}"

# File Storage
FILESYSTEM_DISK=public

# Logging
LOG_CHANNEL=daily
LOG_LEVEL=warning
```

4.2 Generate APP_KEY

WAJIB generate APP_KEY sebelum deployment:

```
php artisan key:generate
```

APP_KEY digunakan untuk enkripsi data sensitif (cookies, session, passwords).

4.3 Konfigurasi Database Production

Untuk database MySQL/PostgreSQL production, edit `.env`:

```
DB_CONNECTION=mysql
DB_HOST=your-database-host.com
DB_PORT=3306
DB_DATABASE=smkn5_profile_db
DB_USERNAME=smkn5_dbuser
DB_PASSWORD=very_strong_password_here
```

---

5. DATABASE & MIGRASI

5.1 Tabel Database

Sistem menggunakan **18 tabel utama**:

Tabel	Deskripsi	Migration File
`users`	Data pengguna (Admin, Redaktur, Jurnalis)	`0001_01_01_000000_create_users_table.php`
`cache`	Cache storage	`0001_01_01_000001_create_cache_table.php`
`jobs`	Queue jobs	`0001_01_01_000002_create_jobs_table.php`
`jurusan`	Program keahlian/jurusan	`2025_12_30_120934_create_jurusan_table.php`
`berita`	Artikel berita	`2025_12_30_120958_create_berita_table.php`
`settings`	Pengaturan website	`2025_12_30_121003_create_settings_table.php`
`extracurriculars`	Data ekstrakurikuler	`2025_12_30_132037_create_extracurriculars_table.php`
`facilities`	Data fasilitas sekolah	`2025_12_30_132037_create_facilities_table.php`
`partners`	Mitra/partner sekolah	`2025_12_30_132038_create_partners_table.php`

Tabel	Deskripsi	Migration File
galleries	Album galeri	2025_12_30_132039_create_galleries_table.php
gallery_images	Foto dalam galeri	2025_12_30_132040_create_gallery_images_table.php
menus	Menu navigasi dinamis	2026_01_03_000001_create_menus_table.php
statistics	Statistik sekolah	2026_01_06_000001_create_statistics_table.php
agendas	Agenda/kegiatan sekolah	2026_01_07_161040_create_agendas_table.php
roles	Role user (Spatie)	2026_01_06_141315_create_permission_tables.php
permissions	Permissions (Spatie)	2026_01_06_141315_create_permission_tables.php
model_has_roles	Pivot user-roles	2026_01_06_141315_create_permission_tables.php
model_has_permissions	Pivot user-permissions	2026_01_06_141315_create_permission_tables.php

5.2 Schema Tabel Utama

Tabel: users

```
CREATE TABLE users (
    id BIGINT UNSIGNED PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(255) NOT NULL,
    email VARCHAR(255) UNIQUE NOT NULL,
    email_verified_at TIMESTAMP NULL,
    password VARCHAR(255) NOT NULL,
    role ENUM('admin', 'redaktur', 'jurnalis') DEFAULT 'jurnalis',
    remember_token VARCHAR(100),
    created_at TIMESTAMP,
    updated_at TIMESTAMP
);
```

Tabel: berita

```
CREATE TABLE berita (
    id BIGINT UNSIGNED PRIMARY KEY AUTO_INCREMENT,
    judul VARCHAR(255) NOT NULL,
    slug VARCHAR(255) UNIQUE NOT NULL,
    konten TEXT NOT NULL,
    gambar VARCHAR(255) NULL,
    user_id BIGINT UNSIGNED,
    status ENUM('draft', 'review', 'published') DEFAULT 'draft',
    is_featured BOOLEAN DEFAULT FALSE,
```

```
    published_at TIMESTAMP NULL,
    created_at TIMESTAMP,
    updated_at TIMESTAMP,
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE SET NULL
);
```

Tabel: jurusan

```
CREATE TABLE jurusan (
    id BIGINT UNSIGNED PRIMARY KEY AUTO_INCREMENT,
    kode VARCHAR(10) UNIQUE NOT NULL,
    nama VARCHAR(255) NOT NULL,
    deskripsi TEXT,
    image VARCHAR(255) NULL,
    durasi_tahun INT DEFAULT 3,
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP,
    updated_at TIMESTAMP
);
```

Tabel: settings

```
CREATE TABLE settings (
    id BIGINT UNSIGNED PRIMARY KEY AUTO_INCREMENT,
    site_name VARCHAR(255),
    site_description TEXT,
    site_logo VARCHAR(255),
    contact_email VARCHAR(255),
    contact_phone VARCHAR(50),
    address TEXT,
    facebook_url VARCHAR(255),
    instagram_url VARCHAR(255),
    twitter_url VARCHAR(255),
    youtube_url VARCHAR(255),
    created_at TIMESTAMP,
    updated_at TIMESTAMP
);
```

5.3 Menjalankan Migrasi

```
# Run all migrations
php artisan migrate

# Rollback last migration
php artisan migrate:rollback

# Reset database (drop all tables + re-migrate)
php artisan migrate:fresh
```

```
# Reset + run seeder
php artisan migrate:fresh --seed
```

5.4 Seeding Data

```
# Run all seeders
php artisan db:seed

# Run specific seeder
php artisan db:seed --class=RolePermissionSeeder
php artisan db:seed --class=UserRoleSeeder
php artisan db:seed --class=JurusanSeeder
```

---

6. SISTEM AUTENTIKASI & KEAMANAN

6.1 Guard & Provider

Sistem menggunakan **Laravel Auth** default dengan konfigurasi:

File: `config/auth.php`

```
'defaults' => [
    'guard' => 'web',
    'passwords' => 'users',
],

'guards' => [
    'web' => [
        'driver' => 'session',
        'provider' => 'users',
    ],
],

'providers' => [
    'users' => [
        'driver' => 'eloquent',
        'model' => App\Models\User::class,
    ],
],
```

6.2 Filament Authentication

Filament menggunakan **FilamentUser** contract untuk akses panel admin.

File: `app/Models/User.php`

```
```

```php
use Filament\Models\Contracts\FilamentUser;
use Filament\Panel;

class User extends Authenticatable implements FilamentUser
{
    /**
     * Determine if user can access Filament panel
     */
    public function canAccessPanel(Panel $panel): bool
    {
        // Hanya Admin, Redaktur, dan Jurnalis yang bisa akses panel
        return in_array($this->role, ['admin', 'redaktur', 'jurnalis']);
    }
}
```

6.3 Password Hashing

Sistem menggunakan **bcrypt hashing** dengan konfigurasi di `config/hashing.php`:

```php
'driver' => 'bcrypt',
'bcrypt' => [
    'rounds' => env('BCRYPT_ROUNDS', 12),
],
```

6.4 Middleware Protection

Panel Admin Routes:

```
// Auto-protected oleh Filament dengan middleware:
- EncryptCookies
- AuthenticateSession
- VerifyCsrfToken
- Authenticate (redirect jika belum login)
```

Frontend Routes:

```
// Public routes (no authentication required)
Route::get('/', [HomeController::class, 'index'])->name('home');
Route::get('/berita', [BeritaController::class, 'index'])->name('berita.index');
Route::get('/jurusan', [JurusanController::class, 'index'])->name('jurusan.index');
```

6.5 CSRF Protection

Semua form **WAJIB** menyertakan CSRF token:

```
<form method="POST" action="/some-endpoint">
    @csrf
    <!-- Form fields -->
</form>
```

Filament menangani CSRF secara otomatis via `VerifyCsrfToken` middleware.

---

7. ROLE & PERMISSION

7.1 Role-Based Access Control (RBAC)

Sistem menggunakan **Spatie Laravel Permission** dengan 3 role utama:

```
┌─────────────────────────────────────────────────────┐
│                    USER ROLES                       │
├──────────────┬──────────────────┬──────────────────┤
│    ADMIN     │    REDAKTUR      │     JURNALIS     │
├──────────────┼──────────────────┼──────────────────┤
│ Full Access  │ Review Berita    │ Write Berita     │
│ Manage Users │ Publish Berita   │ Edit Own Berita  │
│ Manage All   │ View All Berita  │ View Own Berita  │
│ Resources    │ (review+publish) │ (all status)     │
└──────────────┴──────────────────┴──────────────────┘
```

7.2 Permission Matrix

Permission	Admin	Redaktur	Jurnalis
User Management			

Permission	Admin	Redaktur	Jurnalis
View Users	✅	❌	❌
Create User	✅	❌	❌
Edit User	✅	❌	❌
Delete User	✅	❌	❌
Berita Management			
View All Berita	✅	✅ (review+published)	❌
View Own Berita	✅	✅	✅
Create Berita	✅	✅	✅
Edit All Berita	✅	✅ (review+published)	❌
Edit Own Berita	✅	✅	✅
Delete All Berita	✅	✅	❌
Delete Own Berita	✅	✅	✅
Publish Berita	✅	✅	❌
Set Featured	✅	✅	❌
Content Management			
Manage Jurusan	✅	❌	❌
Manage Settings	✅	❌	❌
Manage Galleries	✅	✅	❌
Manage Facilities	✅	❌	❌
Manage Partners	✅	❌	❌
Manage Menus	✅	❌	❌
Manage Statistics	✅	❌	❌

7.3 Berita Workflow

```
┌─────────────────────────────────────────────────────┐
│           BERITA PUBLISHING WORKFLOW                │
└─────────────────────────────────────────────────────┘

JURNALIS:
  [Create Berita] → Status: DRAFT
         ↓
  [Submit for Review] → Status: REVIEW
         ↓
         ↓ (Cannot publish directly)
         ↓
REDAKTUR:
```

```
        ↓
[Review Berita] → Status: REVIEW
        ↓
[Approve & Publish] → Status: PUBLISHED
        ↓
        ↓
[Public Website] ← Only "published" berita shown
```

Status Enum:

- `draft` → Draft awal (visible: Jurnalis pemilik, Admin)
- `review` → Menunggu review (visible: Jurnalis pemilik, Redaktur, Admin)
- `published` → Sudah terbit (visible: Public + all roles)

7.4 Eloquent Query Scope per Role

File: `app/Filament/Resources/Beritas/BeritaResource.php`

```php
public static function getEloquentQuery(): Builder
{
    $query = parent::getEloquentQuery();
    $user = auth()->user();

    // JURNALIS: Hanya lihat berita sendiri (semua status)
    if ($user->role === 'jurnalis') {
        return $query->where('user_id', $user->id);
    }

    // REDAKTUR: Hanya lihat berita 'review' dan 'published'
    if ($user->role === 'redaktur') {
        return $query->whereIn('status', ['review', 'published']);
    }

    // ADMIN: Lihat semua berita
    return $query;
}
```

7.5 Policy-Based Authorization

File: `app/Policies/BeritaPolicy.php`

```php
class BeritaPolicy
{
    public function viewAny(User $user): bool
    {
        return in_array($user->role, ['admin', 'redaktur', 'jurnalis']);
    }

    public function create(User $user): bool
    {
        return in_array($user->role, ['admin', 'redaktur', 'jurnalis']);
```

```
    }

    public function update(User $user, Berita $berita): bool
    {
        // Admin & Redaktur: Edit all berita
        if (in_array($user->role, ['admin', 'redaktur'])) {
            return true;
        }

        // Jurnalis: Hanya edit berita sendiri
        return $berita->user_id === $user->id;
    }

    public function delete(User $user, Berita $berita): bool
    {
        // Admin: Delete all
        if ($user->role === 'admin') return true;

        // Redaktur: Delete all
        if ($user->role === 'redaktur') return true;

        // Jurnalis: Delete own berita only
        return $berita->user_id === $user->id;
    }
}
```

---

8. FITUR FRONTEND

8.1 Halaman Publik

Halaman	Route	Controller	Deskripsi
Homepage	`/`	`HomeController@index`	Berita terbaru, jurusan, statistik, agenda
Berita List	`/berita`	`BeritaController@index`	Daftar berita terbit
Berita Detail	`/berita/{slug}`	`BeritaController@show`	Detail berita
Jurusan List	`/jurusan`	`JurusanController@index`	Daftar program keahlian
Jurusan Detail	`/jurusan/{slug}`	`JurusanController@show`	Detail jurusan
Tentang	`/profil/tentang`	`ProfileController@tentang`	Profil, visi-misi, sejarah
Unit	`/profil/unit`	`ProfileController@unit`	Unit sekolah
Struktur	`/profil/struktur`	`ProfileController@struktur`	Struktur organisasi
Ekstrakurikuler	`/kesiswaan/ekskul`	Static Route	Daftar ekstrakurikuler

8.2 Components (Blade)

File: `resources/views/components/`

Navbar (`x-navbar`)

- **Lokasi:** `resources/views/components/navbar.blade.php`
- **Fitur:**
  - Scroll-aware (transparent → solid on scroll)
  - Responsive mobile menu
  - Dynamic menu dari database (Model: `Menu`)
  - Dark mode toggle (optional)

Footer (`x-footer`)

- **Lokasi:** `resources/views/components/footer.blade.php`
- **Fitur:**
  - Contact info dari `settings` table
  - Social media links
  - Copyright info

8.3 Layout Template

File: `resources/views/layouts/app.blade.php`

```
<!DOCTYPE html>
<html lang="id" class="scroll-smooth scroll-pt-20">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>@yield('title', 'SMKN 5 Samarinda')</title>

    <link rel="icon" href="{{ asset('images/logo-smk-utama.png') }}">
    @vite(['resources/css/app.css', 'resources/js/app.js'])
    @stack('styles')
</head>
<body class="bg-gray-50">
    <x-navbar />
    <div class="h-20"></div> <!-- Navbar spacer -->

    <main>
        @yield('content')
    </main>

    <x-footer />
    @stack('scripts')
</body>
</html>
```

Cara Pakai di View:

```
@extends('layouts.app')

@section('title', 'Judul Halaman')
```

```
@section('content')
    <!-- Content here -->
@endsection

@push('scripts')
    <script>
        // Custom scripts
    </script>
@endpush
```

8.4 Asset Management

Public Assets:

- Logo: `public/images/logo-smk-utama.png`
- Hero Images: `public/images/drone-smk-*.webp`
- Pimpinan Photos: `public/assets/images/pimpinan/*.webp`

Storage (User Uploads):

- Berita Images: `storage/app/public/berita/`
- Gallery Images: `storage/app/public/galleries/`
- Facility Images: `storage/app/public/facilities/`

Symbolic Link:

```
php artisan storage:link
```

Membuat symlink: `public/storage` → `storage/app/public`

---

9. PANEL ADMIN (FILAMENT)

9.1 Akses Panel

URL: `http://your-domain.com/admin`

Login:

- Email: `admin@smkn5samarinda.sch.id`
- Password: `Admin@2026` (ganti setelah instalasi!)

9.2 Dashboard

File: `app/Filament/Widgets/StatsOverview.php`

Widget per Role:

Admin:

- Total Berita
- Total User

- Total Jurusan

Redaktur:

- Menunggu Review
- Sudah Terbit
- Total Berita (yang visible)

Jurnalis:

- Tulisan Saya
- Sudah Terbit
- Menunggu Review

9.3 Resources (CRUD Modules)

Berita Resource

- **Lokasi:** `app/Filament/Resources/Beritas/BeritaResource.php`
- **Model:** `App\Models\Berita`
- **Form Fields:**
  - Judul (text)
  - Slug (auto-generated)
  - Konten (rich editor)
  - Gambar (image upload)
  - Status (select: draft/review/published)
  - Is Featured (toggle)
  - Published At (datetime)
- **Table Columns:**
  - Gambar (thumbnail)
  - Judul
  - Penulis (user relation)
  - Status (badge)
  - Featured (icon)
  - Tanggal Publish
- **Filters:**
  - Status (draft/review/published)
  - Penulis (user)
  - Tanggal (date range)

User Resource

- **Lokasi:** `app/Filament/Resources/Users/UserResource.php`
- **Model:** `App\Models\User`
- **Access:** Admin only
- **Form Fields:**
  - Name
  - Email
  - Password
  - Role (select: admin/redaktur/jurnalis)
- **Table Columns:**

- Name
- Email
- Role (badge)
- Created At

Jurusan Resource

- **Lokasi:** `app/Filament/Resources/Jurusans/JurusanResource.php`
- **Model:** `App\Models\Jurusan`
- **Access:** Admin only
- **Form Fields:**
  - Kode (text, max 10)
  - Nama
  - Deskripsi (textarea)
  - Image (image upload)
  - Durasi Tahun (number)
  - Is Active (toggle)

Setting Resource

- **Lokasi:** `app/Filament/Resources/Settings/SettingResource.php`
- **Model:** `App\Models\Setting`
- **Access:** Admin only
- **Form Fields:**
  - Site Name
  - Site Description
  - Site Logo
  - Contact Email
  - Contact Phone
  - Address
  - Social Media URLs (Facebook, Instagram, Twitter, YouTube)

Gallery Resource

- **Lokasi:** `app/Filament/Resources/Galleries/GalleryResource.php`
- **Model:** `App\Models\Gallery`
- **Relations:**
  - HasMany: `GalleryImage`
- **Form:**
  - Title
  - Description
  - Repeater: Gallery Images (upload multiple)

Facility Resource

- **Lokasi:** `app/Filament/Resources/Facilities/FacilityResource.php`
- **Model:** `App\Models\Facility`
- **Form:**
  - Name
  - Description

- Image
- Is Available (toggle)

Partner Resource

- **Lokasi:** `app/Filament/Resources/Partners/PartnerResource.php`
- **Model:** `App\Models\Partner`
- **Form:**
  - Name
  - Logo
  - Website URL
  - Description

9.4 Branding

File: `app/Providers/Filament/AdminPanelProvider.php`

```
->favicon(asset('images/logo-smk-utama.png'))
->brandName(fn () => new HtmlString('
    <div style="display: flex; align-items: center; gap: 12px;">
        <img src="'.asset('images/logo-smk-utama.png').'"
             alt="Logo SMKN 5"
             style="height: 40px; width: auto;" />
        <span style="font-weight: 700; color: white;">
            SMK NEGERI 5
        </span>
    </div>
'))
->colors([
    'primary' => '#1e5494', // Biru SMKN 5
])
```

---

10. ROUTING & CONTROLLERS

10.1 Frontend Routes

File: `routes/web.php`

```
// Homepage
Route::get('/', [HomeController::class, 'index'])->name('home');

// Berita
Route::prefix('berita')->group(function () {
    Route::get('/', [BeritaController::class, 'index'])->name('berita.index');
    Route::get('/{slug}', [BeritaController::class, 'show'])->name('berita.show');
});

// Jurusan
Route::prefix('jurusan')->group(function () {
    Route::get('/', [JurusanController::class, 'index'])->name('jurusan.index');
```

```
    Route::get('/{slug}', [JurusanController::class, 'show'])->name('jurusan.show');
});

// Profile
Route::prefix('profil')->group(function () {
    Route::get('/tentang', [ProfileController::class, 'tentang'])-
>name('profil.tentang');
    Route::get('/unit', [ProfileController::class, 'unit'])->name('profil.unit');
    Route::get('/struktur', [ProfileController::class, 'struktur'])-
>name('profil.struktur');
});

// Ekstrakurikuler (Static)
Route::get('/kesiswaan/ekskul', function () {
    // Static data or from Extracurricular model
    return view('kesiswaan.ekskul');
})->name('ekskul.index');
```

10.2 Controllers

HomeController

```
namespace App\Http\Controllers;

class HomeController extends Controller
{
    public function index()
    {
        // Featured news (1)
        $featuredNews = Berita::where('status', 'published')
            ->where('is_featured', true)
            ->first();

        // Latest 7 news (excluding featured)
        $latestNews = Berita::where('status', 'published')
            ->when($featuredNews, fn($q) => $q->where('id', '!=', $featuredNews-
>id))
            ->orderBy('published_at', 'desc')
            ->take(7)
            ->get();

        // Combine for slider
        $beritaTerbaru = collect([$featuredNews])->merge($latestNews);

        // Active jurusan
        $jurusan = Jurusan::where('is_active', true)->get();

        // Statistics
        $statistics = Statistic::orderBy('order')->get();

        // Upcoming agendas
        $agendas = Agenda::active()->upcoming()->get();
```

```
        return view('home', compact('beritaTerbaru', 'jurusan', 'statistics',
'agendas'));
    }
}
```

BeritaController

```
namespace App\Http\Controllers;

class BeritaController extends Controller
{
    public function index()
    {
        $berita = Berita::where('status', 'published')
            ->orderBy('published_at', 'desc')
            ->paginate(12);

        return view('berita.index', compact('berita'));
    }

    public function show($slug)
    {
        $berita = Berita::where('slug', $slug)
            ->where('status', 'published')
            ->firstOrFail();

        // Related news (same category or random)
        $related = Berita::where('status', 'published')
            ->where('id', '!=', $berita->id)
            ->inRandomOrder()
            ->limit(3)
            ->get();

        return view('berita.detail', compact('berita', 'related'));
    }
}
```

10.3 Filament Routes

Auto-Registered:

- `/admin` → Login page
- `/admin/login` → Login form
- `/admin/dashboard` → Dashboard
- `/admin/beritas` → Berita CRUD
- `/admin/users` → User management (Admin only)
- `/admin/jurusans` → Jurusan CRUD
- `/admin/settings` → Site settings

Middleware Stack:

- `EncryptCookies`

- `AuthenticateSession`
- `VerifyCsrfToken`
- `Authenticate` (redirect to /admin/login if not authenticated)

---

11. MODELS & RELATIONSHIPS

11.1 User Model

File: `app/Models/User.php`

```
class User extends Authenticatable implements FilamentUser
{
    use HasFactory, Notifiable, HasRoles;

    protected $fillable = ['name', 'email', 'password', 'role'];
    protected $hidden = ['password', 'remember_token'];
    protected $casts = [
        'email_verified_at' => 'datetime',
        'password' => 'hashed',
    ];

    // Filament Access
    public function canAccessPanel(Panel $panel): bool
    {
        return in_array($this->role, ['admin', 'redaktur', 'jurnalis']);
    }

    // Relationships
    public function berita()
    {
        return $this->hasMany(Berita::class);
    }
}
```

11.2 Berita Model

File: `app/Models/Berita.php`

```
class Berita extends Model
{
    protected $table = 'berita';
    protected $fillable = [
        'judul', 'slug', 'konten', 'gambar',
        'user_id', 'status', 'published_at', 'is_featured'
    ];
    protected $casts = [
        'published_at' => 'datetime',
        'is_featured' => 'boolean',
    ];

    // Auto-generate slug
```

```
    protected static function boot()
    {
        parent::boot();

        static::creating(function ($berita) {
            if (empty($berita->slug)) {
                $berita->slug = Str::slug($berita->judul);
            }
        });

        // Only 1 featured news at a time
        static::saving(function ($berita) {
            if ($berita->is_featured) {
                static::where('id', '!=', $berita->id)
                    ->update(['is_featured' => false]);
            }
        });
    }

    // Relationships
    public function user()
    {
        return $this->belongsTo(User::class);
    }
}
```

11.3 Relationship Diagram

```
User (1) ──────── (∞) Berita
  │
  └── HasMany: berita

Berita (∞) ──────── (1) User
  │
  └── BelongsTo: user

Gallery (1) ──────── (∞) GalleryImage
  │
  └── HasMany: images

GalleryImage (∞) ──────── (1) Gallery
  │
  └── BelongsTo: gallery
```

---

12. PANDUAN INSTALASI

12.1 Requirement

- **PHP:** 8.2 atau lebih tinggi
- **Composer:** 2.x
- **Node.js:** 18.x atau lebih tinggi

- **NPM/Yarn:** Latest
- **Database:** SQLite (dev) / MySQL 8.0+ atau PostgreSQL 13+ (production)
- **Web Server:** Apache/Nginx (production)

12.2 Instalasi Development

```
# 1. Clone repository
git clone https://github.com/your-org/profile-smkn5.git
cd profile-smkn5

# 2. Install PHP dependencies
composer install

# 3. Install NPM dependencies
npm install

# 4. Copy environment file
copy .env.example .env

# 5. Generate application key
php artisan key:generate

# 6. Create SQLite database (Windows)
type nul > database\database.sqlite

# 7. Run migrations + seeders
php artisan migrate:fresh --seed

# 8. Create storage symlink
php artisan storage:link

# 9. Build frontend assets
npm run build

# 10. Start development server
php artisan serve
```

Akses:

- Frontend: `http://localhost:8000`
- Admin Panel: `http://localhost:8000/admin`

12.3 Development dengan Vite Hot Reload

```
# Terminal 1: PHP Server
php artisan serve

# Terminal 2: Vite Dev Server
npm run dev
```

12.4 Composer Scripts

```
# Setup otomatis (install + migrate + build)
composer run setup

# Development mode (concurrent server+queue+logs+vite)
composer run dev

# Run tests
composer run test
```

---

13. PANDUAN DEPLOYMENT

13.1 Deployment ke Shared Hosting

Persiapan File

```
# 1. Build production assets
npm run build

# 2. Optimize autoloader
composer install --optimize-autoloader --no-dev

# 3. Cache config
php artisan config:cache
php artisan route:cache
php artisan view:cache
```

Upload ke Server

1. Upload semua file **KECUALI**:
    - `/node_modules`
    - `/tests`
    - `/.git`
    - `/.env` (buat baru di server)
2. **Public Root:**
    - Set document root ke folder `/public`
    - Atau copy isi `public/` ke `public_html/`
3. **File .env Production:**

```
APP_ENV=production
APP_DEBUG=false
APP_URL=https://smkn5samarinda.sch.id

DB_CONNECTION=mysql
DB_HOST=localhost
```

```
DB_DATABASE=smkn5_profile
DB_USERNAME=smkn5_user
DB_PASSWORD=your_db_password
```

4. **Set Permissions:**

```
chmod -R 755 storage
chmod -R 755 bootstrap/cache
```

5. **Generate Key (jika belum):**

```
php artisan key:generate
```

6. **Run Migrations:**

```
php artisan migrate --force
php artisan db:seed --force
```

7. **Create Storage Link:**

```
php artisan storage:link
```

13.2 Deployment ke VPS (Ubuntu)

Install Stack (LEMP)

```
# Update system
sudo apt update && sudo apt upgrade -y

# Install Nginx
sudo apt install nginx -y

# Install PHP 8.2
sudo add-apt-repository ppa:ondrej/php
sudo apt update
sudo apt install php8.2-fpm php8.2-cli php8.2-mysql php8.2-mbstring \
  php8.2-xml php8.2-bcmath php8.2-curl php8.2-zip php8.2-gd -y

# Install MySQL
sudo apt install mysql-server -y
sudo mysql_secure_installation

# Install Composer
curl -sS https://getcomposer.org/installer | php
sudo mv composer.phar /usr/local/bin/composer
```

```
# Install Node.js 18
curl -fsSL https://deb.nodesource.com/setup_18.x | sudo -E bash -
sudo apt install nodejs -y
```

Nginx Configuration

File: `/etc/nginx/sites-available/smkn5`

```
server {
    listen 80;
    server_name smkn5samarinda.sch.id;
    root /var/www/smkn5/public;

    add_header X-Frame-Options "SAMEORIGIN";
    add_header X-Content-Type-Options "nosniff";

    index index.php;

    charset utf-8;

    location / {
        try_files $uri $uri/ /index.php?$query_string;
    }

    location = /favicon.ico { access_log off; log_not_found off; }
    location = /robots.txt  { access_log off; log_not_found off; }

    error_page 404 /index.php;

    location ~ \.php$ {
        fastcgi_pass  unix:/var/run/php/php8.2-fpm.sock;
        fastcgi_param SCRIPT_FILENAME $realpath_root$fastcgi_script_name;
        include fastcgi_params;
    }

    location ~ /\.(?!well-known).* {
        deny all;
    }
}
```

Enable Site:

```
sudo ln -s /etc/nginx/sites-available/smkn5 /etc/nginx/sites-enabled/
sudo nginx -t
sudo systemctl restart nginx
```

Deploy Application

```
# Clone repository
cd /var/www
sudo git clone https://github.com/your-org/profile-smkn5.git smkn5

# Set permissions
sudo chown -R www-data:www-data /var/www/smkn5
sudo chmod -R 755 /var/www/smkn5/storage
sudo chmod -R 755 /var/www/smkn5/bootstrap/cache

# Install dependencies
cd /var/www/smkn5
composer install --optimize-autoloader --no-dev
npm install && npm run build

# Configure environment
cp .env.example .env
nano .env  # Edit database credentials

# Setup database
php artisan key:generate
php artisan migrate --force
php artisan db:seed --force
php artisan storage:link

# Optimize
php artisan config:cache
php artisan route:cache
php artisan view:cache
```

SSL Certificate (Let's Encrypt)

```
sudo apt install certbot python3-certbot-nginx -y
sudo certbot --nginx -d smkn5samarinda.sch.id
```

13.3 Post-Deployment Checklist

- [ ] `.env` file configured correctly
- [ ] `APP_KEY` generated
- [ ] Database migrated and seeded
- [ ] Storage symlink created
- [ ] File permissions set (755 for storage)
- [ ] Production caches cleared (`config:cache`, `route:cache`, `view:cache`)
- [ ] Change default admin password
- [ ] Test all pages (frontend + admin)
- [ ] SSL certificate installed
- [ ] Backup strategy configured

---

14. KEAMANAN & BEST PRACTICES

14.1 Security Checklist

Environment Security

```
# WAJIB set di production
APP_ENV=production
APP_DEBUG=false        # Jangan tampilkan error details
SESSION_SECURE_COOKIE=true  # HTTPS only
```

Password Policy

- **Minimum:** 8 karakter
- **Recommended:** 12+ karakter, kombinasi huruf besar/kecil/angka/simbol
- **Ganti password default** setelah instalasi

File Upload Security

```
// Validasi file upload (sudah diterapkan di Filament)
'gambar' => 'nullable|image|mimes:jpeg,png,jpg,webp|max:2048', // Max 2MB
```

CSRF Protection

- Semua form **WAJIB** pakai `@csrf`
- Filament auto-handle CSRF via middleware

SQL Injection Prevention

- **JANGAN** raw queries: ~~`DB::raw("SELECT * FROM users WHERE id = $id")`~~
- **GUNAKAN** Eloquent ORM atau parameter binding:

```
// ☑ AMAN
User::where('id', $id)->first();

// ☑ AMAN (Query Builder)
DB::table('users')->where('id', $id)->first();

// ✖ TIDAK AMAN
DB::select("SELECT * FROM users WHERE id = $id");
```

XSS Prevention

- Blade auto-escape: `{{ $variable }}` (aman)
- Raw HTML: `{!! $html !!}` (gunakan hanya jika HTML trusted)
- Sanitize user input sebelum simpan

14.2 Permissions Best Practices

```
# Development (Windows)
# File permissions umumnya tidak masalah

# Production (Linux)
# Owner: www-data (web server user)
sudo chown -R www-data:www-data /path/to/smkn5

# Storage & Cache: Writable (755)
sudo chmod -R 755 storage
sudo chmod -R 755 bootstrap/cache

# Jangan chmod 777! (security risk)
```

14.3 Backup Strategy

Database Backup

```
# Development (SQLite)
copy  database\database.sqlite  backup\db_backup_2026-02-09.sqlite

# Production (MySQL)
mysqldump -u smkn5_user -p smkn5_profile > backup_2026-02-09.sql

# Automated (Cron Job)
0 2 * * * mysqldump -u user -ppassword database > /backups/db_$(date +\%Y\%m\%d).sql
```

File Backup

```
# Backup uploads
tar  -czf  storage_backup_2026-02-09.tar.gz  storage/app/public/

# Full backup (exclude vendor, node_modules)
tar  --exclude='vendor'  --exclude='node_modules'  \
    -czf  smkn5_full_backup_2026-02-09.tar.gz  /path/to/smkn5
```

14.4 Monitoring & Logging

```
// Check logs
tail -f storage/logs/laravel.log
```

```
// Log levels
Log::emergency($message);
Log::alert($message);
Log::critical($message);
Log::error($message);
Log::warning($message);
Log::notice($message);
Log::info($message);
Log::debug($message);
```

Production Log Config:

```
LOG_CHANNEL=daily  # Rotate logs daily
LOG_LEVEL=warning  # Only log warning and above
```

---

15. TROUBLESHOOTING

15.1 Common Errors

Error: "No application encryption key has been specified"

Solusi:

```
php artisan key:generate
```

Error: "SQLSTATE[HY000]: General error: 1 no such table"

Solusi:

```
# Run migrations
php artisan migrate

# Or reset database
php artisan migrate:fresh --seed
```

Error: "The stream or file could not be opened"

Solusi:

```
# Linux
sudo chmod -R 755 storage
sudo chown -R www-data:www-data storage
```

```
# Windows
# Check folder write permissions
```

Error: "Class 'Spatie\Permission\Models\Role' not found"

Solusi:

```
# Install missing package
composer require spatie/laravel-permission

# Publish migration
php artisan vendor:publish --provider="Spatie\Permission\PermissionServiceProvider"

# Run migration
php artisan migrate
```

Error: "Vite manifest not found"

Solusi:

```
# Development
npm run dev

# Production
npm run build
```

Error: "The POST method is not supported"

Penyebab: Missing `@csrf` token

Solusi:

```
<form method="POST">
    @csrf
    <!-- form fields -->
</form>
```

15.2 Performance Issues

Slow Page Load

```
# Cache configuration
php artisan config:cache
php artisan route:cache
php artisan view:cache
```

```
# Optimize autoloader
composer dump-autoload --optimize

# Clear cache (if needed)
php artisan cache:clear
php artisan config:clear
php artisan route:clear
php artisan view:clear
```

Slow Queries

```
// Enable query log
DB::enableQueryLog();

// Your code here

// Get queries
dd(DB::getQueryLog());
```

Optimize dengan Index:

```
// Migration
Schema::table('berita', function (Blueprint $table) {
    $table->index('status');
    $table->index('published_at');
});
```

15.3 Database Issues

Reset Database

```
# Fresh migrations
php artisan migrate:fresh

# Fresh + seeder
php artisan migrate:fresh --seed
```

Check Database Connection

```
php artisan tinker
>>> DB::connection()->getPdo();
```

---

16. MAINTENANCE & UPDATE

16.1 Regular Maintenance

```
# Clear old logs (manual)
rm  storage/logs/laravel-*.log

# Optimize application
php artisan optimize

# Clear expired cache
php  artisan  cache:prune-stale-tags
```

16.2 Update Dependencies

```
# Update PHP packages
composer update

# Update NPM packages
npm update

# Check outdated packages
composer outdated
npm outdated
```

16.3 Laravel Upgrade

```
# Check current version
php artisan --version

# Upgrade Laravel (composer.json)
composer require laravel/framework:^13.0

# Update autoloader
composer update

# Clear all caches
php artisan optimize:clear

# Re-cache
php artisan optimize
```

16.4 Filament Upgrade

```
# Upgrade Filament
composer require filament/filament:^4.4

# Run upgrade command
php artisan filament:upgrade
```

```
# Clear cache
php artisan optimize:clear
```

16.5 Backup Before Update

SELALU backup sebelum update major version:

```
# 1. Backup database
mysqldump -u user -p database > backup_pre_update.sql

# 2. Backup files
tar -czf smkn5_backup_pre_update.tar.gz /path/to/smkn5

# 3. Test di development dulu
# 4. Baru deploy ke production
```

📞 LINK

Official Website:

- https://smkn5smr.sch.id
